

Two Ways to Sample Library Visits

AP Statistics · Unit 3 · Topic 3.9 · TEACHER KEY

CED TETHER

- **UNC-3.N** Determine parameters of a sampling distribution for a difference in sample proportions. **[Skill 3.B]**
- **UNC-3.N.1** For a categorical variable, when randomly sampling from two independent populations with proportions p_1 and p_2 , the sampling distribution of the difference in sample proportions has mean $\mu = p_1 - p_2$.
- **UNC-3.N.2** If sampling without replacement and sample sizes are less than 10% of population sizes, the difference is negligible.
- **UNC-3.O** Determine whether a sampling distribution for a difference of sample proportions can be described as approximately normal. **[Skill 3.C]**
- **UNC-3.O.1** The sampling distribution of the difference in sample proportions will have an approximate normal distribution provided sample sizes are large enough: $n_1 p_1 \geq 10$, $n_1(1 - p_1) \geq 10$, $n_2 p_2 \geq 10$, $n_2(1 - p_2) \geq 10$.
- **UNC-3.P** Interpret probabilities and parameters for a sampling distribution for a difference in proportions. **[Skill 4.B]**
- **UNC-3.P.1** Parameters for a sampling distribution for a difference of proportions should be interpreted using appropriate units and within the context of specific populations.

Essential question: When is the independent two-proportion sampling model warranted, and what changes for paired observations?

DOK-3 skill: 4.B · **FRQ pattern:** independent-vs-paired-which-sampling-distribution-model-is-warranted

DOK rationale: Part (c) compares independent and repeated-person sampling plans, evaluates the useful and unsupported parts of competing claims, and explains how a mismatched model can misstate uncertainty.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- What random object is selected in Plan I, and what is selected in Plan P?
- Which four expected counts establish the normal shape for Plan I?
- Does using 120 twice make the two results independent?
- What information about within-cardholder changes disappears if the plans are treated alike?

[WARN] LOOK FOR

- Treating equal sample sizes as evidence of independence.
- Subtracting the two variance terms because the proportions are subtracted.
- Calling Plan P unusable instead of identifying the need for a paired analysis.

SCORING PART (c) — E / P / I

Expected elements

- **judgment-1** (required) — Applies the independent model to Plan I using separate random selection and passing conditions.
- **judgment-2** (required) — Recognizes the useful within-person comparison in Plan P while explaining its linked responses.
- **judgment-3** (required) — Explains that equal sample sizes do not imply independence and transferring the SD/tail can misstate uncertainty.

E	Meets all three checks with correct contextual reasoning; equivalent wording is acceptable.
P	Shows at least one substantive correct check, but has an omission or error that prevents E.
I	Shows none of the three checks with substantive correct reasoning; a choice alone is insufficient.

Common mistakes

- Treating equal sample sizes as an independence condition
- Subtracting the variance contributions because the proportions are subtracted
- Using the independent-sample normal probability for the same 120 people
- Calling the paired design invalid rather than changing the analysis

[STAR] THE PROBLEM — annotated

A library studies self-checkout before and after a redesign. Records give $p_B = 0.42$ among 4,000 before visits and $p_A = 0.54$ among 5,000 after visits.

Plan I: Draw an SRS of 120 before visits and a separate SRS of 150 after visits; do not track cardholders.

Plan P: Measure the same 120 cardholders before and after.

Sora: Separate selection warrants Plan I's independent model; 0.0350 means a difference at most 0.01 is unusual.

Mateo: Pairing controls person differences, so Plan P may help. Because both proportions use 120 people, Plan I's model applies to P too.

For Plan I, the four expected counts are 81, 69, 50.4, and 69.6. A normal calculator gives the lower-tail probability 0.0350.

Sampling plans

Plan	Before	After	Repeated
I	SRS 120	SRS 150	no
P	120	same 120	yes

FIRST TAKE — one sentence before you work the parts

Does using the same people twice seem different from choosing two separate groups? Give one reason.

Key: Not graded. Sora matches the model; Mateo values a sound design but transfers the wrong model.

[BOOK] THE VARIANCE FORMULA REQUIRES INDEPENDENT SAMPLES (keep on the page)

For independent samples, $\mu_{\hat{p}_1 - \hat{p}_2} = p_1 - p_2$ and $\sigma_{\hat{p}_1 - \hat{p}_2} = \sqrt{p_1(1 - p_1)/n_1 + p_2(1 - p_2)/n_2}$ when each sample is below 10% of its population. Approximate normality requires all four expected counts to be at least 10. Repeated measurements on the same people are paired, not independent; their association belongs in the variability.

DOK 1 (a) For after minus before, identify the parameter and statistic and state Plan I's sampling-distribution mean.

Key: The parameter $p_A - p_B$ is the after-minus-before population difference; the statistic is $\hat{p}_A - \hat{p}_B$. Its mean is $0.54 - 0.42 = 0.12$.

DOK 2 (b) For Plan I, verify both 10% conditions and use the four supplied counts to check normality. Calculate the SD and interpret the supplied probability $P(\hat{p}_A - \hat{p}_B \leq 0.01) = 0.0350$.

Key: Both 10% checks pass: $150 < 500$ and $120 < 400$. All four supplied counts exceed 10. The SD is $\sqrt{0.54(0.46)/150 + 0.42(0.58)/120} = 0.0607124$. About 3.5% of independent sample pairs would show an after-minus-before difference at most 0.01 when the population difference is 0.12.

DOK 3 (c) In 3–4 sentences, explain what Sora and Mateo each get right and wrong. Use how the samples are selected to decide where the independent model applies. Explain why transferring its SD and probability to the repeated people could mislead a reader.

Key: Sora's model fits Plan I: separate SRSs and passing conditions support the independent formula. Mateo is right that repeated people can help compare within-person changes. However, their linked re-

sponses are dependent, even with equal sample sizes. Transferring the independent SD and tail probability to Plan P ignores those links and can misstate uncertainty.